

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026 — ANSWER KEY (Version 1)

Biology — Class IX | Paper: SSC-I

Syllabus: Ch 1 — The Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle
| Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec
7.1-7.6)

SECTION A — Multiple Choice Questions: Answers & Explanations

Q1. Correct Answer: C) Aristotle 1 mark

The Greek philosopher **Aristotle** was the first person who classified living organisms, well before Linnaeus introduced binomial nomenclature. Later classifiers such as **Abu Usama Aljahiz** (700s), **Andrea Caesalpino**, **John Ray**, and **Tournefort** built on this early work.

Q2. Correct Answer: B) Ernst Haeckel 1 mark

The German scientist **Ernst Haeckel** proposed the **three-kingdom classification system**, adding the kingdom **Protista** to accommodate organisms like *Euglena* that show both plant-like and animal-like characteristics, separating unicellular microscopic organisms from multicellular ones.

Q3. Correct Answer: C) Nine 1 mark

Centrioles are hollow, open-ended cylindrical structures found in animal cells, existing in pairs near the nuclear envelope. Each centriole consists of **nine triplets of microtubules**, and they help form the **spindle apparatus** during cell division.

Q4. Correct Answer: D) Auxins 1 mark

In **cloning by tissue culture**, a scraped tissue sample is placed in an agar growth medium containing nutrients and the plant hormone **auxins**, allowing the sample to develop into tiny plantlets that are later planted into compost.

Q5. Correct Answer: C) Skeletal 1 mark

The **skeletal system** (bones, cartilage, joints, tendons, ligaments) maintains homeostasis by **regulating the level of calcium and other minerals** in the blood, storing or releasing them from bones as needed. It also produces blood cells.

Q6. Correct Answer: D) Friedrich Miescher 1 mark

The Swiss physician **Friedrich Miescher** isolated a whitish material from the nucleus of human pus cells and named it "**nuclein**". Based on its acidic properties, it was later renamed **nucleic acid**.

Q7. Correct Answer: B) 2 nm 1 mark

The **Watson-Crick model** of DNA states that the diameter of the double helix is thoroughly uniform and is approximately **2 nm**, regardless of the base sequence.

Q8. Correct Answer: C) 36-38°C 1 mark

Each enzyme has a specific **optimum temperature**. For human enzymes, this range is **36°C to 38°C**; temperatures far above this damage the enzyme's globular structure and active site, causing **denaturation**.

Q9. Correct Answer: C) 8.00 1 mark

Trypsin, active in the alkaline environment of the small intestine, performs best at pH **8.00**, while **pepsin** (in the acidic stomach) performs best at pH **2.00**.

Q10. Correct Answer: D) 36 1 mark

The complete oxidation of one glucose molecule during **cellular respiration** generates a net total of approximately **36 ATP** molecules, which the cell uses as its “energy currency”.

Q11. Correct Answer: B) ratio and proportion 1 mark

In the **biological method**, once data (facts/information) is collected, it is organized using **tables and graphs**. To predict outcomes on the basis of this data is called **analysis**, which is done by means of **ratio and proportion**.

Q12. Correct Answer: C) 220 1 mark

Around **220 types of cells** are identified in the human body, all developing from a single **zygote** through the process of **differentiation**. The zygote is the most basic **stem cell**.

SECTION B – Short Question Answers

2(i) – Woese's three domain classification criteria 3 marks

The **THREE** criteria Carl Woese used to classify organisms into Archaea, Bacteria and Eukarya:

(1) the sequence of nucleotides in the **rRNA (ribosomal RNA)** of the cell 1

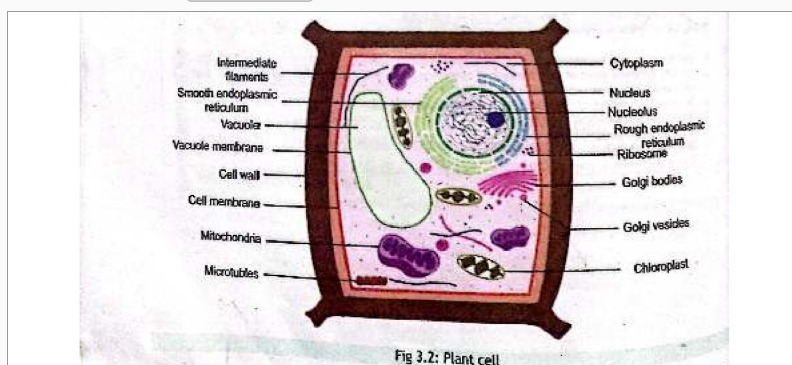
(2) the cell's **membrane lipid structure** 1

(3) the cell's **sensitivity to antibiotics** 1

OR Two-kingdom classification system: the two kingdoms used are **Plantae** and **Animalia** 1.

Kingdom Plantae included the autotrophs, as well as **bacteria, fungi and algae** 1. Kingdom Animalia included organisms that depend on autotrophs or other heterotrophs 1.

2(ii) – Label the plant cell figure 3 marks



A = Nucleus 0.75 **B = Chloroplast** 0.75 **C = Vacuole** 0.75 **D = Cell wall** 0.75

OR Cell specialization (differentiation) is the process by which daughter cells formed by mitosis alter in size, structure, metabolic activities and physiological responses to perform their specific role in the body 1. A **root hair cell** is a modified epidermal cell – a long, tubular outgrowth containing a large vacuole and many mitochondria 1. Its elongated structure increases surface area for absorbing water and minerals from the soil, and the many mitochondria supply energy for this active absorption 1.

2(iii) – Haematopoietic stem cells / vacuole types 3 marks

A **haematopoietic stem cell** is a type of stem cell that gives rise to blood-related cell types 1. Any **FOUR** of: **red blood cells, white blood cells, platelets, muscle cells, nerve cells, liver cells** 2 (0.5 each).

OR (1) **Temporary vacuoles** of animal cells – store water and food substances 1. (2) **Contractile vacuoles** of freshwater organisms (e.g. Amoeba, sponges) – collect and pump out extra water and wastes 1. (3) **Large central vacuole** of plant cells – contains cell sap and maintains **turgor pressure** 1.

2(iv) — STEAM activities: plant cell / animal cell slide preparation 3 marks

(1) Place a small piece of onion skin in a drop of water on a slide and cover with a cover slip **1**. (2) Observe under the microscope, first under the **low power objective** **1**. (3) Then observe under the **high power objective** and draw the diagram **1**.

OR (1) Gently pass the broad end of a toothpick on the inner side of the cheek **1**. (2) Place the material in a drop of **methylene blue solution** on a slide and cover with a cover slip **1**. (3) Observe under low power then high power objective and draw the diagram **1**.

2(v) — Human chromosome numbers & meiosis / G1 phase 3 marks

Human somatic cells are **diploid = 46 chromosomes** **1**; gametes (egg, sperm) are **haploid = 23 chromosomes** **1**. Meiosis halves the chromosome number so that fertilization ($23 + 23$) restores 46 in the zygote; if gametes were formed by mitosis instead, they would carry 46 chromosomes each, and the number would double every generation **1**.

OR During **G1 phase**, the cell grows, increases the number of organelles, and forms proteins **1**, preparing itself for the S phase **0.5**. It specifically forms the **enzymes and nucleotides** needed for DNA replication, so that once the S phase begins, all raw materials are ready for accurate copying of the chromosomes **1.5**.

2(vi) — Skeletal system / Urinary system 3 marks

Organs: bones, cartilage, joints, tendons, and ligaments **1.5**. **Role in homeostasis:** regulates the level of **calcium and other minerals** in the blood, by storing or releasing them from bones; also supports/protects the body and produces blood cells **1.5**.

OR Organs: kidneys, ureters, urinary bladder, and urethra **1.5**. **Role in homeostasis:** removes excess water, salts and waste products from the blood and body; regulates the levels of water, salts, H^+ and wastes in the blood, controlling pH **1.5**.

2(vii) — Friedrich Miescher / iodine test starch vs glycogen 3 marks

Friedrich Miescher (Swiss physician) **1** isolated a whitish material from the nucleus of human pus cells and named it "**nuclein**" **1**. Based on its acidic properties, it was later renamed **nucleic acid** **1**.

OR Iodine test: add a few drops of iodine solution to the sample. Starch + iodine → **dark blue-black** colour **1**. If the sample were **glycogen** instead, it would give a **RED** colour, not blue-black **1**. Both are storage polysaccharides but differ structurally, so the colour reaction distinguishes one from the other **1**.

2(viii) — RNA structure / monosaccharide open-chain vs ring 3 marks

RNA is **single stranded** **1**. Its pentose sugar is **ribose**; the base that replaces thymine is **uracil** **1**. The nitrogen bases are described as "overhung" because, since RNA is single stranded, they are **not bonded to a complementary strand** as they are in DNA **1**.

OR Monosaccharides exist in an **open chain structure** when in crystalline (solid) form **1**, and convert to a **ring structure** when dissolved in water **1**. The ring structure (e.g. of ribose) is the form more commonly found in the nucleotides of nucleic acids **1**.

2(ix) — Pepsin/trypsin pH / low vs high temperature 3 marks

Pepsin (stomach) optimum pH = **2.00** (acidic) **1**; **Trypsin** (small intestine) optimum pH = **8.00** (alkaline) **1**. Reason: pepsin works in the acidic stomach (HCl present) while trypsin works in the alkaline small intestine; each enzyme's active site is shaped to function best at the specific pH of its working environment **1**.

OR At **low temperature**, enzymes have insufficient activation energy available, so they become inactive — but their globular structure and active site are **not damaged**, so activity returns when normal temperature is restored **1.5**. At very **high temperature**, atoms vibrate vigorously, permanently damaging the globular structure and active site (**denaturation**) — this cannot be reversed **1.5**.

2(x) — ATP yield & daily usage / STEAM amylase-temperature 3 marks

Complete oxidation of one glucose molecule → net **36 ATP** molecules 1.5 . An average human cell uses **100 to 150 moles** of ATP per day 1.5 .

OR At **40°C (optimum)**, amylase worked fastest and broke down all the starch completely before the iodine test, so **no** blue-black colour appeared 1 . At **0°C**, insufficient activation energy meant the reaction was very slow, so starch remained (blue-black appears) 1 . At **60°C/80°C**, high temperature denatured the enzyme's active site, so it could not break down the starch (blue-black appears) 1 .

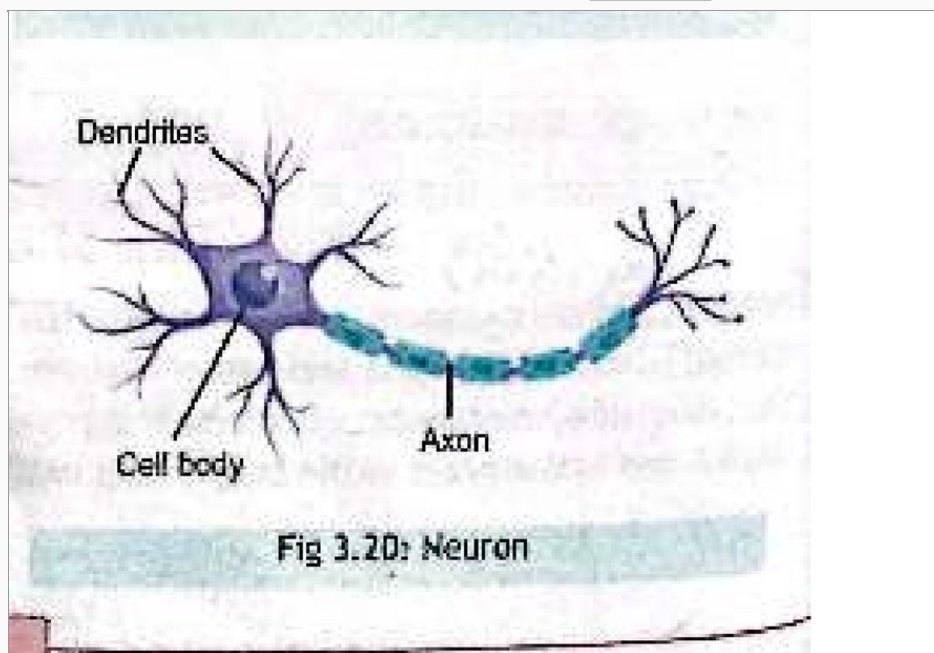
2(xi) — Fig 7.2 breakdown vs synthesis / chewing-amylase-HCl-pepsin 3 marks

Breakdown reaction (Fig 7.2a): substrate binds active site → **ES complex** → catalyzed to **EP complex** → breaks into TWO separate products + free enzyme 1.5 . **Synthesis reaction (Fig 7.2b):** TWO separate substrates bind the active site together → ES complex → EP complex → combine into ONE product + free enzyme 1.5 .

OR Chewing food for longer produces more starch-digesting enzyme **amylase**, which mixes into the food and helps break down starch 1 . Chewing also triggers the production of **HCl** in the stomach 1 , which lowers the stomach's pH to activate the enzyme **pepsin**, which then breaks down protein 1 .

SECTION C — Long Question Answers

3(i) — Neuron diagram / structure-function comparison 5 marks



Correct diagram with three parts labelled 2 . **Cell body** - contains the nucleus, controls the neuron's activities 1 . **Dendrites** - conduct nerve impulses *TO* the cell body 1 . **Axon** - conducts messages *AWAY FROM* the cell body 1 .

OR A **neuron's** cell body, dendrites and axon are suited to coordination, allowing it to communicate with far-away cells of the body 1.7 . A **liver cell** is round with a prominent nucleus and abundant organelles, suiting its role as the most metabolically active cell (glycogen storage, detoxification, clotting-protein production, RBC recycling) 1.7 . A **mesophyll cell** contains a large number of chloroplasts, suiting its role of carrying out photosynthesis in the leaf 1.6 .

3(ii) — Shoot/root system / 4-organ-system comparison table 5 marks

Shoot system (above ground): includes **vegetative** parts (leaves, stems) and **reproductive** parts (flowers, fruits) **1.5**. Functions: leaves perform photosynthesis; stems support the plant and transport substances; the shoot system absorbs light needed for photosynthesis **1**. **Root system** (underground): anchors the plant into the ground **0.5**; absorbs water and minerals, and serves as a storage site for food **2**.

OR

System	Main Organs	Homeostatic Role
Skeletal	Bones, cartilage, joints, tendons, ligaments	Regulates calcium/mineral levels in blood
Urinary	Kidneys, ureters, bladder, urethra	Regulates water, salts, H ⁺ , wastes, pH
Lymphatic	Lymph, lymph nodes, lymph vessels	Maintains tissue fluid homeostasis
Muscular	Skeletal, cardiac, smooth muscles	Heat production for temperature regulation

(any 4 correct systems with organs + role, 1.25 marks each) **5**

3(iii) — DNA base percentage calculation / three protein functions 5 marks

Given **cytosine = 34%**. Since C≡G always pair equally, **guanine = 34%** **1.5**. Remaining bases (A+T) = 100–68 = 32%; since A=T always, **adenine = thymine = 16%** each **2**. This follows **Chargaff's/Watson-Crick base-pairing rule**: adenine always equals thymine (2 hydrogen bonds), and cytosine always equals guanine (3 hydrogen bonds) **1.5**.

OR (a) **Cell membrane proteins** act as channel and carrier proteins, and as **antigens** that define blood groups **1.7**. (b) **Fibrinogen** – a blood clotting protein found in plasma **1.7**. (c) **Hormones** – chemical messengers regulating processes such as growth, reproduction, digestion, and glucose levels (any TWO) **1.6**.

3(iv) — Graph interpretation / STEAM 7.2 pepsin-Biuret experiment 5 marks

Fig 7.3 (temperature): rate increases with temperature up to the optimum (36–38°C for human enzymes), then decreases sharply due to **denaturation** **1.7**. **Fig 7.4 (pH)**: each enzyme (pepsin/trypsin) has its own optimum pH (2.00 / 8.00); the rate decreases sharply on either side due to denaturation at extreme pH **1.7**. **Fig 7.5 (substrate concentration)**: rate increases proportionally with substrate concentration, then plateaus once all active sites are occupied — **saturation of active sites** **1.6**.

OR Tube A (neutral pH): pepsin cannot break down protein at neutral pH (not its optimum), so pink/purple colour appears with Biuret solution (protein still present) **1.7**. **Tube B (acidic pH)**: pepsin works at its optimum (~pH 2–3), completely breaking down the protein, so **no** pink/purple colour appears (protein absent) **1.7**. **Tube C (alkaline pH)**: pepsin is denatured at alkaline pH and cannot break down protein, so pink/purple colour appears again (protein still present) **1.6**.

— End of Answer Key —